Sephora Products & Reviews Analysis

Tools: Python (Pandas), SQL (PostgreSQL), Power BI

1. Executive Summary

This project demonstrates a full-scale Data Engineering and Analytics pipeline. I processed raw data containing Sephora products and customer reviews to uncover pricing strategies, brand performance, and consumer sentiment trends.

2. Data Engineering Pipeline (Python)

The first phase focused on **ETL (Extract, Transform, Load)** to ensure data quality and consistency.

Key Procedures:

- **Data Consolidation:** Merged multiple review files into a single 1M+ row dataset using glob.
- **Advanced Imputation:** Mapped missing prices and brand names from the product table to the reviews table to ensure 100% data integrity for financial analysis.
- **Feature Engineering:**
    - Created discount_percentage and discount_buckets.
    - Segmented products by popularity and rating_category using quantiles.
    - Defined price_range bins for better dashboard filtering.

3. SQL Business Intelligence Insights

After loading the cleaned data into **PostgreSQL**, I ran a series of business-critical queries to analyze the market.

A. Pricing & Promotion Analysis

We analyzed how Sephora handles discounts across different price segments and categories.

- **Metric:** Catalog-wide discount penetration and average discount depth.
- **Insight:** Identifying which categories are most "promotion-heavy."

Percent Products on Sale:

	percent_products_on_sale numeric
1	3.18

Average Discount per Category:

	primary_category text	avg_discount_percentage numeric
1	Tools & Brushes	50.00
2	Mini Size	46.61
3	Makeup	46.27
4	Skincare	40.46
5	Men	40.00
6	Fragrance	37.84
7	Bath & Body	35.00
8	Hair	30.83

Observation: Category Tools&Brushes has the highest % of discounts.

B. Brand & Popularity Metrics

By analyzing loves_count, we identified market leaders and the correlation between price and demand.

- **Metric:** Top 10 Most Loved Brands.
- **Metric:** Price vs. Popularity correlation.

SQL Result for "Top 10 Most Loved Brands":

	brand_name text	total_loves_count numeric	avg_loves_per_product numeric
1	SEPHORA COLLECTION	12530142	35596.99
2	Fenty Beauty by Rihanna	9369443	114261.50
3	Anastasia Beverly Hills	7941142	83590.97
4	Charlotte Tilbury	7725055	78030.86
5	The Ordinary	7449373	137951.35
6	NARS	7442525	111082.46
7	Dior	6230997	45816.15
8	Urban Decay	5730058	97119.63
9	Rare Beauty by Selena Gom...	5724687	143117.18
10	tarte	4995414	38132.93

Observation: This table highlights the brands with the strongest community engagement.

C. Sentiment & Review Quality

We deep-dived into customer satisfaction to find "at-risk" categories.

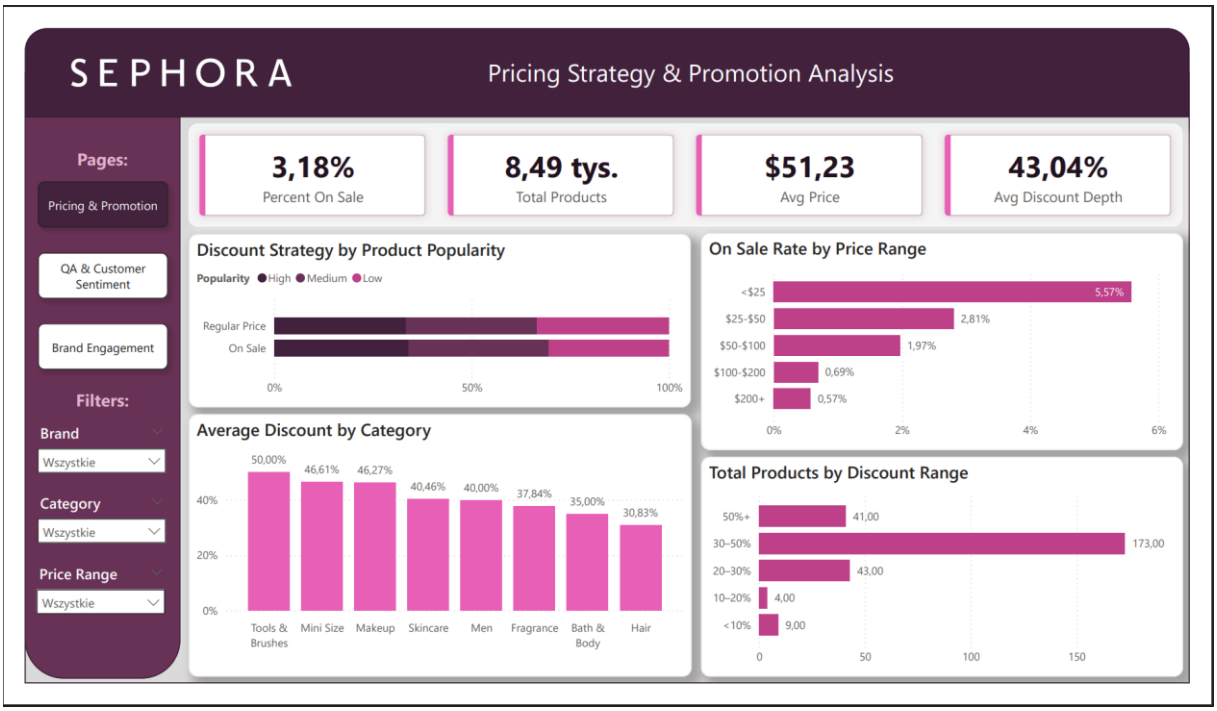
- **Metric:** Correlation between review length and helpfulness.
- **Metric:** Low-rated products percentage per category.

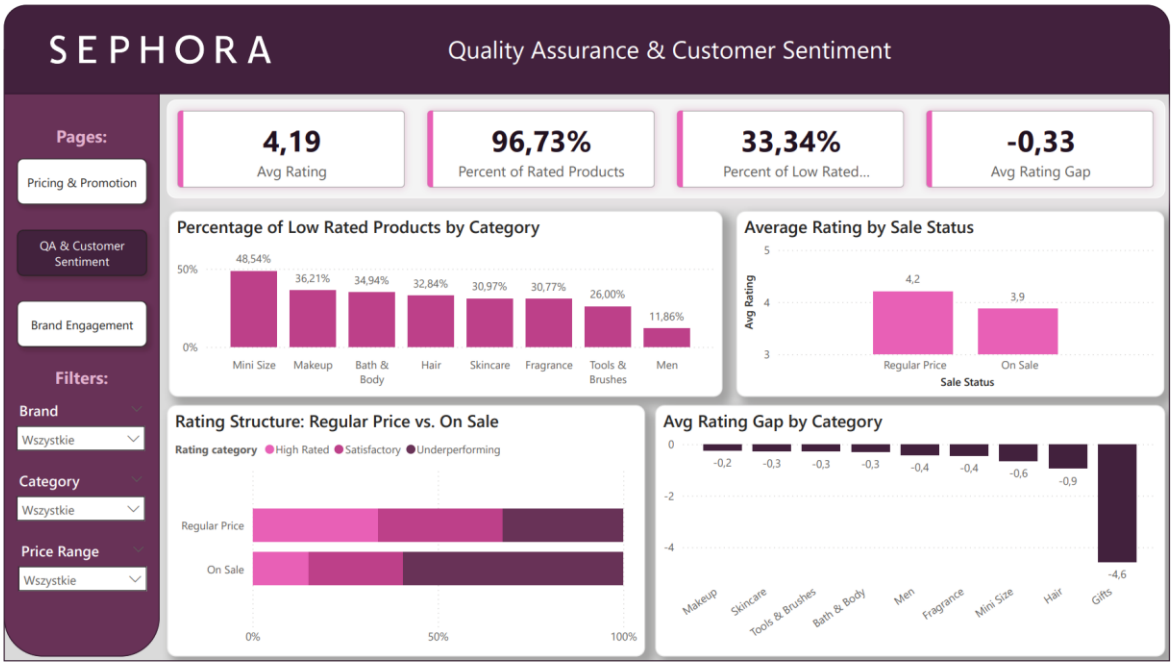
Low Rated Products in Categories:

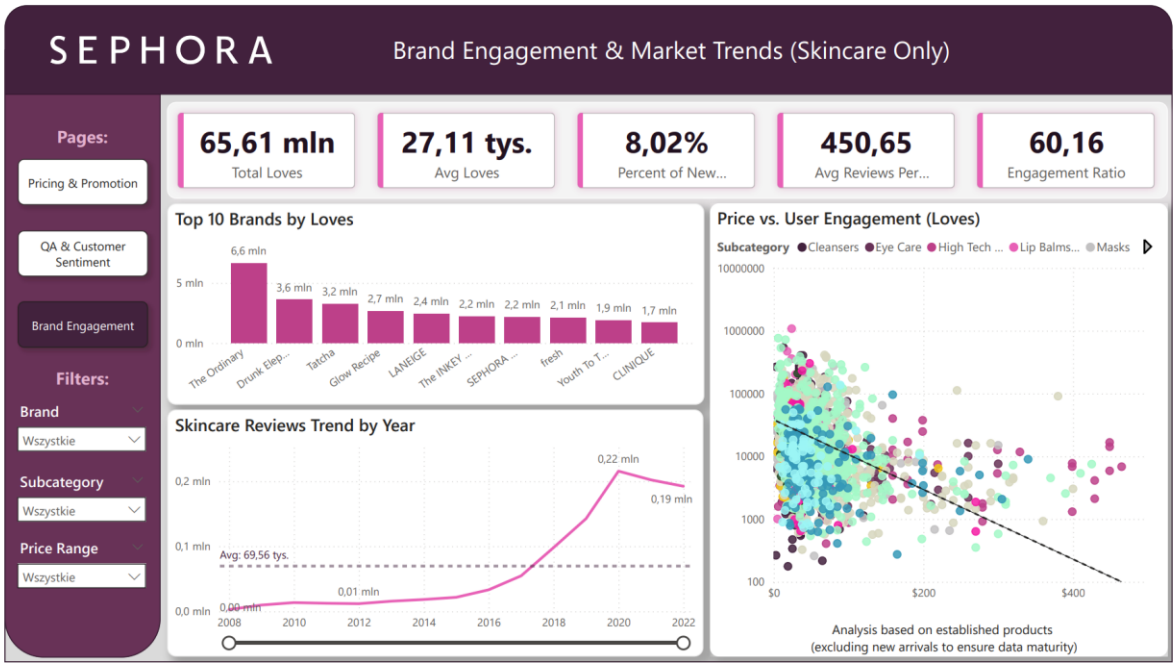
	primary_category text	secondary_category text	total_products bigint	low_rated_products bigint	low_rated_pct numeric
1	Bath & Body	Self Tanners	11	2	18.00
2	Mini Size	Value & Gift Sets	6	1	16.00
3	Makeup	Nail	52	6	11.00
4	Hair	Mini Size	28	3	10.00
5	Makeup	Accessories	45	4	8.00
6	Tools & Brushes	Beauty Tools	23	2	8.00
7	Skincare	High Tech Tools	80	7	8.00
8	Fragrance	Candles & Home Scents	262	18	6.00
9	Hair	Value & Gift Sets	91	6	6.00
10	Mini Size	Fragrance	15	1	6.00

Observation: This identifies which sub-categories might have quality control issues.

4. Data Visualization: Power BI Dashboard




SEPHORA
Quality Assurance & Customer Sentiment
Pages:
Pricing & Promotion
QA & Customer Sentiment
Brand Engagement
Filters:
Brand
Wszystkie
Category
Wszystkie
Price Range
Wszystkie
4,19
Avg Rating
96,73%
Percent of Rated Products
33,34%
Percent of Low Rated...
-0,33
Avg Rating Gap
Percentage of Low Rated Products by Category
50%
0%
48,54%
36,21%
34,94%
32,84%
30,97%
30,77%
26,00%
11,86%
Mini Size
Makeup
Bath & Body
Hair
Skincare
Fragrance
Tools & Brushes
Men
Average Rating by Sale Status
Avg Rating
5
4
3
4,2
3,9
Regular Price
On Sale
Sale Status
Rating Structure: Regular Price vs. On Sale
Rating category High Rated Satisfactory Underperforming
Regular Price
On Sale
0%
50%
100%
Avg Rating Gap by Category
0
-2
-4
-0,2
-0,3
-0,3
-0,3
-0,4
-0,4
-0,6
-0,9
-4,6
Makeup
Skincare
Tools & Brushes
Bath & Body
Men
Fragrance
Mini Size
Hair
Gifts


SEPHORA
Brand Engagement & Market Trends (Skincare Only)
Pages:
Pricing & Promotion
QA & Customer Sentiment
Brand Engagement
Filters:
Brand
Wszystkie
Subcategory
Wszystkie
Price Range
Wszystkie
65,61 mln
Total Loves
27,11 tys.
Avg Loves
8,02%
Percent of New...
450,65
Avg Reviews Per...
60,16
Engagement Ratio
Top 10 Brands by Loves
5 mln
0 mln
6,6 mln
3,6 mln
3,2 mln
2,7 mln
2,4 mln
2,2 mln
2,2 mln
2,1 mln
1,9 mln
1,7 mln
The Ordinary
Drunk Elep...
Tatcha
Glow Recipe
LANEIGE
The INKEY ...
SEPHORA ...
fresh
Youth To T...
CLINIQUE
Skincare Reviews Trend by Year
0,2 mln
0,1 mln
0,0 mln
Avg: 69,56 tys.
0,22 mln
0,19 mln
0,01 mln
0,00 mln
2008
2010
2012
2014
2016
2018
2020
2022
Price vs. User Engagement (Loves)
Subcategory Cleansers Eye Care High Tech ... Lip Balms... Masks
10000000
1000000
100000
10000
1000
100
$0
$200
$400
Analysis based on established products
(excluding new arrivals to ensure data maturity)